

Appendix: Artifact Description/Artifact Evaluation

Artifact Description (AD)

1 Overview of Contributions and Artifacts

1.1 Paper’s Main Contributions

- C_1 A multi-head attention mechanism that combines local gate embeddings with a compact global circuit summary in the critic; the actor retains local embeddings for per-gate rewrite selection (Sec. III-C).
- C_2 A dual-tower GNN (the “Guider”) trained with margin ranking loss on expert trajectories. Given the rewrite rules applicable to a selected gate, the Guider injects a soft logit bias toward its top-10 ranking while keeping all applicable actions available (Sec. III-D).
- C_3 A recursive RL-OAC scheme that optimizes circuit segments, re-optimizes boundary windows after recomposition to retain cross-segment opportunities, and scales to circuits with up to 27,883 gates (Sec. III-E).

1.2 Computational Artifacts

- A_1 GUIDERQ source code, locked models and rule sets, benchmark circuits, and Functional/Reusable smoke-test scripts. <https://doi.org/10.5281/zenodo.19827721>
Current Chameleon image metadata and launch instructions: <https://github.com/quantumlearner/sc26ae-515>

Artifact ID	Contributions Supported	Related Paper Elements
A_1	C_1, C_2, C_3	Tables IV–V, Figures 5–6

2 Artifact Identification

2.1 Computational Artifact A_1

Relation To Contributions

Artifact A_1 contains the GUIDERQ implementation: the multi-head actor-critic (C_1), the Guider ranking network with its IPC-based inference daemon (C_2), and OAC optimization paths (C_3). The `latest_ae/` directory bundles 24 Quartz-suite and 38 MQTBench benchmark circuits (62 unique names), ECC rule sets for the IBM, Nam, and TDG gate sets, and scripts for training, evaluation, and analysis.

Expected Results

Reference results appear in Tables IV–V and Figures 5–6. On the Quartz suite (Table IV), geometric-mean gate-count reductions are 33.0% for IBM, 34.2% for Nam, and 25.1% for TDG. On the 38 MQTBench circuits (Table V, Nam gate set), the geometric-mean reduction over Qiskit-O3 is 10.8%, compared to 5.1% for Quarl. The Functional/Reusable smoke-test verifies deterministic execution: multi-head PPO collects trajectories and updates policy parameters, the Guider returns top-10 rankings to bias online inference, and RL-OAC completes partitioning, segment optimization, boundary recomposition, and QASM generation.

Expected Reproduction Time (in Minutes)

On the pre-built Chameleon image, environment and action-space checks require under one minute, and smoke-all takes up to ten minutes total. The four concurrent A100 lanes cover all TDG, IBM, and Nam multi-head, Guider, and Recursive PPO RL-OAC paths. For contribution-specific checks, run `smoke-multihead` all for C_1 , `smoke-guider` all for C_2 , and `smoke-recursive-rl-oac` all for C_3 . Child process launches require up to 60 seconds each.

A full from-scratch GUIDERQ reproduction requires 72 GPU-hours to pre-train the three gate-set models (3 gate sets $\times$ 4 A100s $\times$ 6 hours) and approximately 2 GPU-hours to prepare ranking data and train the three Guiders. For online optimization, circuits with ≤ 100 , 101–1,000, 1,001–10,000, and $> 10,000$ input gates receive 1, 2, 3, and 4 hours on one A100, respectively. Under the five-seed protocol, Table IV requires 660 GPU-hours, Table V requires 515 GPU-hours, and the two additional Figure 5–6 ablation arms require 50 GPU-hours. The complete GUIDERQ pipeline thus requires approximately 1,299 GPU-hours. Matched Quarl online runs add 1,175 GPU-hours prior to baseline pre-training.

Because the resulting 2,474 GPU-hours exceed standard AE time limits, the artifact provides pre-trained checkpoints, locked Guider models, and a reduced fixed-seed matrix covering ten TDG circuits from Table IV, the Figure 5–6 ablations, and a paired Table V RL-OAC comparison on `realamprandom_indep_130` (27,880 gates). Execution requires 30.33 GPU-hours, with an ideal four-GPU wall time of 7.58 hours; please reserve up to eight hours for launch overhead, validation, and queue imbalance.

Artifact Setup (incl. Inputs)

Hardware. All software dependencies and the experiment environment are pre-installed in the launch-ready Chameleon image; no environment reconstruction or compilation is required after launch. On Chameleon, please select an x86 node with either NVIDIA A100 PCIe or A100 NVLink GPUs. During GUIDERQ development, experiments ran on servers with 4 $\times$ NVIDIA A100 GPUs (80 GB), Intel Xeon Platinum 8358P CPUs (2.60 GHz, 128 cores), and 512 GB RAM. The released image and parallel AE path were validated on a four-A100 PCIe node.

Software. The artifact is written in Python 3.10 and runs on Linux (tested on kernel 5.15.0) with CUDA ≥ 11.7 (PyTorch and DGL are built against CUDA 11.7; system runtime is CUDA 12.8). Main dependencies:

Package	Version	Package	Version
PyTorch	2.0.1	Quartz	0.1.0
DGL	0.9.1post1	NumPy / Matplotlib	1.26.4 / 3.7.5
Qiskit	2.1.1	Pandas	2.3.3

Environment manifests under `latest_ae/environment/` are provided for auditing and fallback reconstruction.

Datasets / Inputs. Inputs are organized into three groups: (i) the 24 Quartz-suite high-level sources and their TDG inputs under `latest_ae/circuits/`; (ii) six checkpoint-compatible, paper-preprocessed IBM inputs and six Nam inputs, with pre-processing records and tools, in the same directory; (iii) all 38 MQTBench circuits from Table V (41–27,883 gates) under

latest_ae/circuits/mqtbench_table_v/, with source, Qiskit-O3-with-measurement, and Nam PPO-ready representations. The three locked ECC rule sets are under latest_ae/ecc/; preprocessing programs are under latest_ae/preprocessing/.

Installation and Deployment. The current Chameleon image name and launch instructions are at <https://github.com/quantumlearner/sc26ae-515>.

After installation, latest_ae/ae.sh check validates dependencies, checksums, bundled Quartz paths, ECC/actor action dimensions, and locked checkpoint structure.

Artifact Evaluation (AE)

3.1 Computational Artifact A_1

Artifact Setup (incl. Inputs)

(1) Start the prepared Chameleon image [0 GPU-h]. Please obtain the current image name, UUID, site, and flavor from:

```
1 https://github.com/quantumlearner/sc26ae-515
```

(2) Set the environment and artifact root [0 GPU-h]. Please connect to the instance and run:

```
1 cd /home/cc/guiderq/latest_ae
2 export LATEST_AE_PYTHON=/home/cc/quartz-ae/env/bin/python
3 export AE_GPUS="0 1 2 3"
```

(3) Validate the environment and locked assets [0 GPU-h].

```
1 ./ae.sh check
```

Please confirm successful checks for Python, CUDA, manifests, checkpoints, all circuit inputs (24 TDG, six IBM, six Nam, 38 MQTBench), and action dimensions (TDG=3,551, IBM=2,651, Nam=28,317).

Artifact Execution

(1) Run all Functional/Reusable smoke tests [up to 10 min wall time on four A100 GPUs; up to 0.67 GPU-h].

```
1 ./ae.sh smoke-all
```

Please use this default entry to test the entire end-to-end workflow. Please allow up to ten minutes for four concurrent GPU lanes. Upon completion, verified logs and output checkpoints/QASM files will be generated under results/history/smoke-runs/full_parallel_<timestamp>/. This command contains all $C_1/C_2/C_3$.

(Optional) To test individual components separately, please use the following sub-commands:

(a) Test the compact global representation (C_1) separately [approx. 0.05 GPU-h].

```
1 BUDGET_SEC=60 ./ae.sh smoke-multihead all
```

Please use this entry to isolate multi-head PPO trajectory collection, policy updates, and final-checkpoint generation.

(b) Test guided exploration (C_2) separately [approx. 0.05 GPU-h].

```
1 BUDGET_SEC=60 ./ae.sh smoke-guider all
```

Please use this entry to isolate the Guider daemon, QASM/node IPC top-10 response, and soft-bias-assisted multi-head PPO.

(c) Test scalable segment optimization (C_3) separately [approx. 0.15 total GPU-h].

```
1 BUDGET_SEC=60 ./ae.sh smoke-recursive-rl-oac all
```

Please use this entry for recursive partitioning, leaf-circuit PPO, meld-boundary PPO, recomposition, and output QASM validation.

(2) Reproduce Table IV, Figures 5–6, and the long Nam representative [30.33 GPU-h]. Please start the resumable four-GPU sequence in a detached terminal:

```
1 tmux new-session -d -s guiderq-reproduce \
2 'cd /home/cc/guiderq/latest_ae && \
3 mkdir -p results/reproduced && \
4 LATEST_AE_PYTHON=/home/cc/quartz-ae/env/bin/python \
5 AE_GPUS="0 1 2 3" ./ae.sh reproduce-all \
6 --run-root results/reproduced 2>&1 | \
7 tee results/reproduced/reproduce_all.log'
```

The fixed-seed 28-task matrix allocates 30.33 aggregate GPU-hours: the ten-circuit TDG subset of Table IV, the Figure 5–6 ablations, and the paired three-hour RL-OAC comparison on realamprandom_indep_130. The ideal four-GPU wall time is 7.58 hours; please reserve up to eight hours for launch overhead, validation, and queue imbalance. After all tasks finish, this command automatically summarizes the logs and generates all CSV, JSON, table, and figure outputs. Please re-execute the same command to resume completed tasks after an interruption.

Artifact Analysis (incl. Outputs)

Live progress [0 GPU-h]. Please inspect task states with:

```
1 ./ae.sh status --run-root results/reproduced
```

Final result locations [0 GPU-h]. After reproduction completes, please print the automatically generated result paths with:

```
1 ./ae.sh results --run-root results/reproduced
```

Recovery and manual refresh [0 GPU-h]. When recovering an older run missing improvement events, please reconstruct them from preserved raw logs with:

```
1 ./ae.sh extract-events --run-root results/reproduced
```

After event recovery, manual result edits, or an interruption before the final summary, please regenerate outputs with:

```
1 ./ae.sh summarize --run-root results/reproduced
```

Please find all generated tables and figures under results/reproduced/summary/paper/, including table_iv_tdg_reproduced in .tex, vector .pdf, and .png formats; the one-row table_v_realamprandom_reproduced comparison with input, output, and reduction values; and the figure5_convergence and figure6_convergence plots. Please find supporting equal-quality timing analyses under results/reproduced/summary/analysis/, the underlying long Nam CSV at results/reproduced/summary/table_v_long_representatives.csv, and raw commands, logs, events, QASM files, and task status records stored in results/reproduced/tasks/.

A canonical reference run is provided under reference_results/canonical_20260726/, which remains unmodified during execution. It preserves original Table IV and Figures 5–6 outputs and long Nam attempts on realamprandom_indep_130.

Across multiple validation runs during artifact preparation, geometric-mean gate reductions for Table IV consistently reach approximately 32–33% for GUIDERQ versus 30–31% for Quarl. For Table V (realamprandom_indep_130, 27,880 gates), GUIDERQ achieves 9–11% reduction compared to 1–3% for Quarl. For Figures 5 and 6, although random seeds are fixed, intrinsic stochasticity in RL exploration can yield minor per-run variations; we thus provide time-to-same-gate analysis in results/reproduced/summary/analysis/, demonstrating a 3×–5× speedup for GUIDERQ over Quarl. These reproducible results consistently support the main claims of the paper.